PHILIPS AKINBAMI

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EDUCATION

University of Maryland — *College Park, MD*
M.S. in Telecommunications — May 2026
Relevant Coursework: Introduction to Digital Communications Systems, Networks and Protocols I & II, AI Applications for 6G, Cellular Communication Networks, 5G System Design and Evolution.

Howard University — *Washington, DC*
B.S. in Electrical Engineering, GPA 3.70 — May 2024
Relevant Coursework: Wireless Communication, Communications Theory, Electromagnetics, Introduction to Computer Networks.
Honors: $15,000 NSBE Scholarship (academic excellence & leadership) · Dean's List, Fall 2023 & Spring 2024.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MATLAB, Pandas, NumPy, Matplotlib, Excel, Power BI, Git
Networking & Protocols: TCP/IP, IPv4/IPv6, VLANs, STP, SDN, DHCP, DNS, LLDP, SNMP, Routing & Switching
Automation & Analysis Tools: Ansible, NETCONF, RESTCONF, RESTful APIs, Wireshark, tcpdump, GNS3, Juniper Junos
Systems, Cloud & Infrastructure: Linux, Bash/Shell, VMware, Docker, AWS
Certification: JNCIA-DevOps

PROJECTS

MAWI Network Traffic Analyzer | Python, Pandas, Streamlit, PyShark, Wireshark, TCP/IP — GitHub
- Parsed **904,000+ packets** from a MAWI backbone capture and engineered **17 traffic features** covering volume, protocols, endpoint activity, and packet behavior.
- Identified **6,403 TCP retransmissions** and calculated reliability metrics to support large-scale traffic analysis and troubleshooting.
- Built an interactive Streamlit dashboard with KPI summaries, filters, protocol views, retransmission trends, and downloadable feature data.

5G Integration and Test Lab | Open5GS, srsRAN, Docker, Ansible, Python, Linux, Bash — GitHub
- Implemented Open5GS from source using a multi-stage Docker workflow and deployed **5 core network functions**—NRF, UDR, UDM, AUSF, and AMF.
- Configured MongoDB persistence, SBI networking, service discovery, PLMN, TAC, slicing, and NGAP/SCTP in a virtual **5G Standalone** environment.
- Validated heartbeat behavior and **network-function registration** across the deployed core services using container logs and NRF service-discovery records.

Cell Site Energy Analyzer | Python, pandas, Matplotlib — GitHub
- Compared **2.3M** cellular activity records across **10,000** Milan grid cells to identify the highest-demand area, then derived a normalized **24-hour** traffic profile for detailed energy analysis.
- Developed a configurable traffic-to-power and battery model that estimated **47.69 kWh** daily energy consumption for a modeled **10 kWh** battery system.
- Detected a **12-hour risk window** where backup runtime fell below the **4-hour target**, reaching **2.4 hours** at peak traffic, and visualized traffic, power demand and battery resilience in Matplotlib.

WORK EXPERIENCE

Cummins | System Analytics and Modeling Intern — *Columbus, IN · May 2023 – Aug 2023*
- Created Power BI dashboards used by **20+ engineers** to monitor KPIs, improve visibility into operational performance, and support data-driven decision making.
- Processed and consolidated datasets of **~100,000+ records**, improving reporting accuracy and enabling reliable KPI tracking for engineering teams.
- Authored step-by-step documentation supporting the team's migration from MATLAB workflows to Power BI, improving data-visualization consistency.

Cummins | Reliability Engineering Intern — *Columbus, IN · May 2022 – Aug 2022*
- Validated components using MATLAB and Excel-based reliability models across **10+ design evaluation cycles**.
- Documented failure trends and system performance issues, producing **8+ technical reports** for engineering review.
- Designed an onboarding database that streamlined training and improved workflow efficiency for a team of **15+ members**.